A Factorial Benchmark of Voice Cloning and Lip Synchronization Pipelines

for Spanish-Language Talking Head Generation

Author Names | Institution

Conference Name, 2026

THE GAP

Why we need factorial benchmarks

Motivation

Talking head generation combines two AI components:

Voice Cloning (VC)

Generate speech in a target speaker's voice

- XTTS-v2, kNN-VC, OpenVoice, CosyVoice...
- Evaluated in isolation (ClonEval, RVCBench)

Lip Synchronization

Animate face to match audio

- Wav2Lip, SadTalker, VideoReTalking...
- Evaluated in isolation (THEval: 17 models)

The problem: No benchmark evaluates them together in a factorial design.
Practitioners must combine components but have no evidence about interactions.

What Exists vs. What's Missing

Benchmark	VC	Lipsync	Factorial?	Purpose
THEval (2025)	0	17	No	Lipsync quality
ClonEval (2025)	5	0	No	VC quality
RVCBench (2026)	11	0	No	VC robustness
AV-Deepfake1M++ (2025)	5	3	Yes	Deepfake detection
This work	**4**	**3**	**Yes**	**Quality assessment**

First factorial VC × Lipsync quality benchmark on same stimuli

Research Questions

Hypotheses

H1 VC system affects audio quality
H2 Lipsync system affects visual quality
H3 VC × Lipsync interaction exists
H4 Emotion degrades quality
H5 Emotion × Tool interaction
H6 Metrics correlate with human ratings

Design

$4\ \mathrm{VC} \times 3\ \mathrm{Lipsync} \times 2\ \mathrm{Emotions}$

- 5 Spanish-speaking actors
- 2 clips per condition (5s each)
- 10 computational metrics
- PLACEHOLDER Human eval (N=30)

METHOD

4 VC × 3 Lipsync × 2 Emotions × 5 Actors

Source Material

5 Actors, 2 Conditions

Actor	Gender	Neutral condition	Emotional condition
George	M	Reading instructions	Dramatic monologue
Jordi	M	Reading instructions	Dramatic monologue
Lisset	F	Reading instructions	Dramatic monologue
Maisa	F	Reading instructions	Dramatic monologue
Selene	F	Reading instructions	Dramatic monologue

Processing Pipeline

1. **Extract** 2 clips × 5s from each ~60s video → **20 source clips**
2. **Face crop** to 512×512 via MediaPipe (two-pass: detect → smooth 7-frame → crop 1.5× padding)
3. **Extract audio** at 16 kHz mono | **Language:** Spanish

Voice Cloning Systems

System	Approach	Input	Language	Output
XTTS-v2	Text re-synthesis	Whisper ASR → TTS	Spanish (explicit)	Multi-rate
kNN-VC	kNN feature matching	Audio → audio	Agnostic	16 kHz
OpenVoice V2	Tone color transfer	Audio → audio	Agnostic	Multi-rate
CosyVoice 2	Zero-shot TTS	Whisper ASR → TTS	Spanish (native)	22 kHz

- **Text-based** (XTTS-v2, CosyVoice): transcribe source → re-synthesize with reference voice
- **Audio-based** (kNN-VC, OpenVoice): directly convert voice timbre

Result: 20 clips × 4 VC = **80 VC outputs** (100% success)

Lip Synchronization Systems

System	Input type	Method
Wav2Lip	Video + audio	Modify mouth region of input frames
SadTalker	Still image + audio	Generate from 3D motion coefficients
VideoReTalking	Video + audio	DNet + LNet + ENet pipeline
~~MuseTalk~~	~~Video + audio~~	Disabled: mmpose ↔ Python 3.13

Result: 80 VC × 3 lipsync = 240 planned → **232 generated** (96.7%)

- 8 failures: SadTalker face detection errors

All outputs standardized: **512×512, 25 fps, H.264**

Evaluation Metrics (10)

Synchronization (5)

Metric	Measures
LSE-C	Sync confidence (SyncNet proxy)
LSE-D	Sync distance (SyncNet proxy)
AVSu	AV sync, utterance (AV-HuBERT proxy)
AVSm	AV sync, matched vs GT
LMD	Lip landmark distance vs GT

Visual (2) + Audio (3)

Metric	Measures
SSIM	Structural similarity vs GT
CPBD	Sharpness (Laplacian variance)
WavLM sim	Speaker embedding similarity
Mel sim	Spectral similarity
WER	Word error rate (Whisper)

Statistical test: One-way ANOVA per metric × factor, BH FDR at $\alpha = 0.05$

RESULTS

232 videos, 10 metrics, 30 ANOVA tests

The Big Picture: Clean Factorization

VC system controls **audio** quality | Lipsync system controls **visual** quality | Emotion = **no effect**

Factor	Significant effects	Metrics affected	η^2 range
VC system	5 / 10	WavLM, Mel, WER, LSE-C, LSE-D	0.20 – **0.73**
Lipsync system	6 / 10	SSIM, LMD, CPBD, LSE-C, LSE-D, AVSm	0.04 – **0.85**
Emotion	0 / 10	None	< 0.01

11 out of 30 tests significant after FDR correction

All in expected directions: VC → audio, Lipsync → visual

H1: VC System Dominates Audio Metrics

Metric	F	p	η^2
WavLM sim	204.98	<.001	.730
Mel sim	57.69	<.001	.432
LSE-C	42.44	<.001	.358
LSE-D	41.93	<.001	.356
WER	17.58	<.001	.196

VC explains **43–73%** of variance in audio quality metrics

Speaker Similarity by VC

System	WavLM sim
OpenVoice V2	0.934
kNN-VC	0.918
XTTS-v2	0.895
CosyVoice 2	0.673

OpenVoice V2 (audio-to-audio) preserves speaker identity best

H2: Lipsync System Dominates Visual Metrics

ANOVA Results

Metric	F	p	η^2
SSIM	623.44	<.001	.845
LMD	282.64	<.001	.712
CPBD	18.38	<.001	.138

Lipsync explains **71–85%** of variance in visual quality metrics

Visual Quality by Lipsync System

System	SSIM	LMD ↓	CPBD
Wav2Lip	0.920	4.65	91.8
VideoReTalking	0.916	8.46	109.7
SadTalker	0.510	30.87	66.9

Wav2Lip & VideoReTalking preserve visual structure; SadTalker (image-based) diverges

H4: Emotion Has No Effect

Emotion condition (neutral vs. emotional) had **no significant effect** on any of the 10 metrics

Metric	F	p	η^2	Significant?
WavLM	3.29	.071	.014	No
LMD	2.87	.092	.012	No
LSE-D	2.33	.129	.010	No
Mel sim	1.19	.276	.005	No
LSE-C	1.19	.276	.005	No
SSIM	0.69	.407	.003	No
AVSm	0.55	.460	.002	No
WER	0.15	.700	.001	No
AVSu	0.03	.852	<.001	No
CPBD	0.03	.853	<.001	No

H4: Why No Emotion Effect?

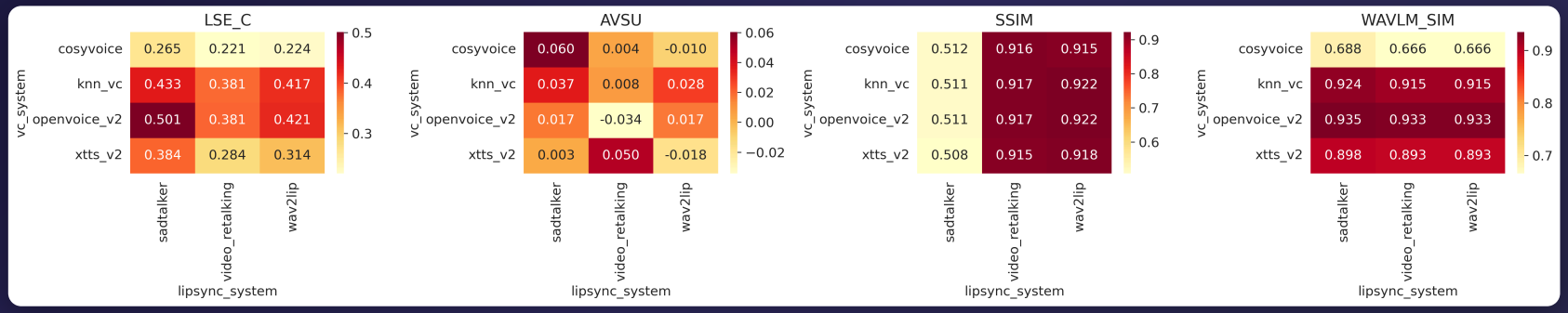
All 10 metrics: $p > .05$, max $\eta^2 = 0.014$ (WavLM)

Three possibilities:

1. **Dramatic monologue ≠ discrete emotions** — unlike ClonEval's categorical emotions (happy, sad, angry), our stimuli use continuous emotional speech
2. **Five-second clips too short** — emotion effects may need longer durations to accumulate measurable degradation
3. **Computational metrics insensitive** — metrics may be genuinely blind to emotion quality; human evaluation needed

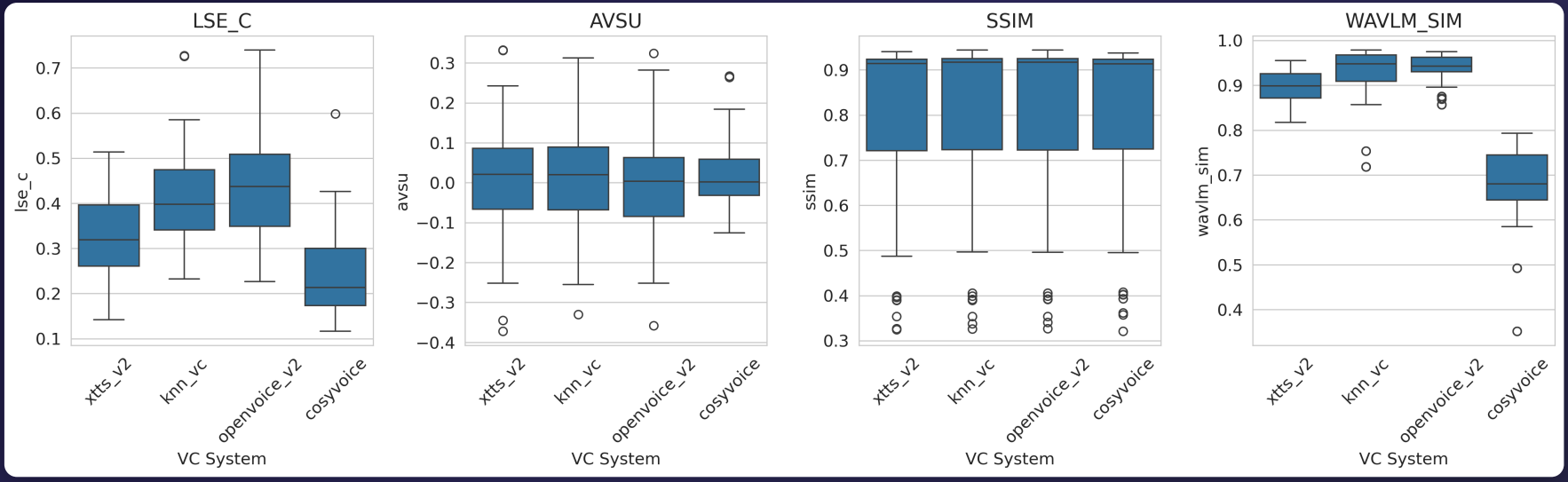
This makes **H6** (metric validity via human correlation) especially important as a next step.

Heatmaps: VC × Lipsync Interaction

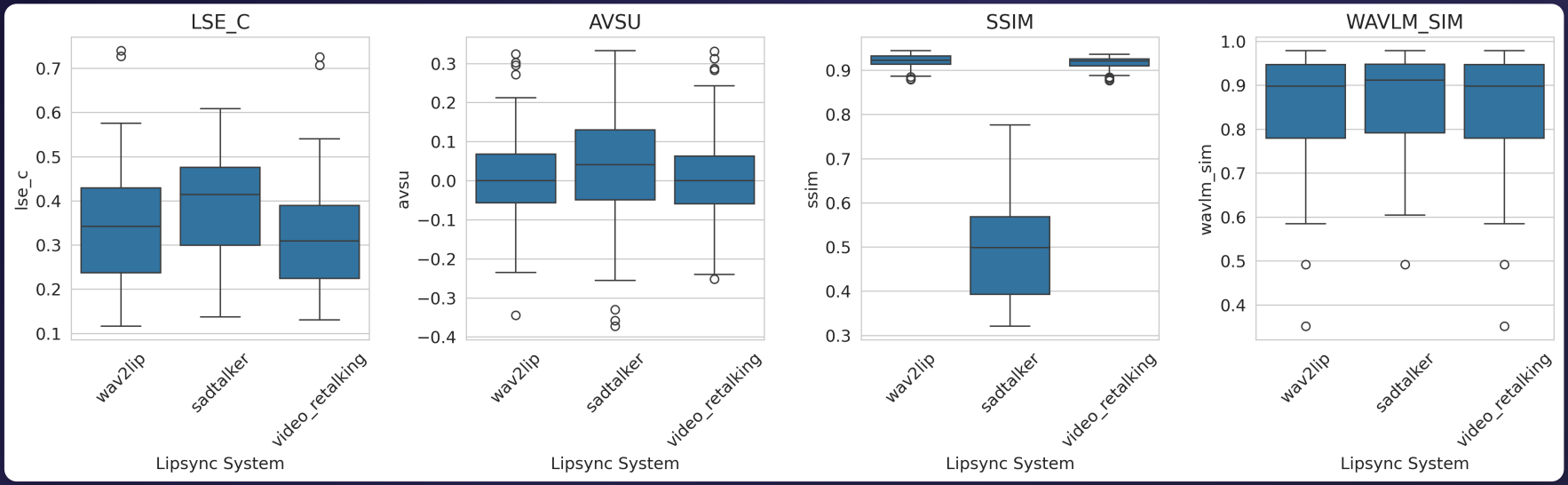


Rows = VC systems, Columns = Lipsync systems. Audio metrics vary by row (VC), visual metrics vary by column (Lipsync).

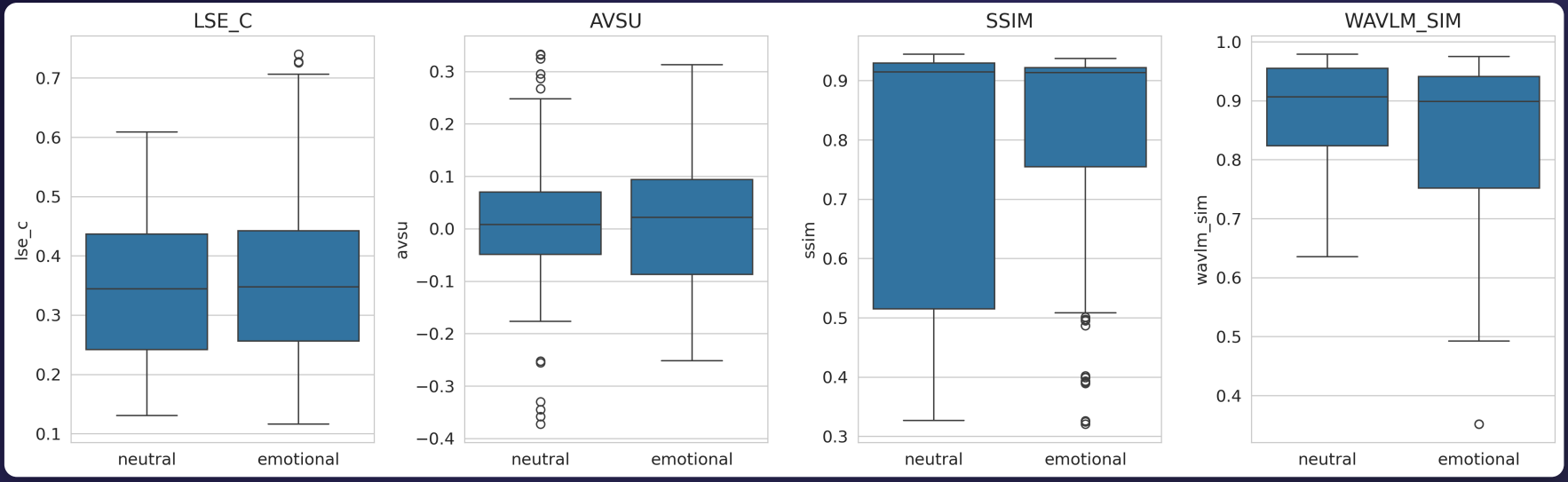
VC System Comparison



Lipsync System Comparison



Emotion Comparison



DISCUSSION

What this means for practitioners

Key Takeaways

Independence

VC and lipsync affect **non-overlapping** metric domains:

- VC → audio (η^2 = 0.20–0.73)
- Lipsync → visual (η^2 = 0.14–0.85)
- Cross-domain: η^2 < 0.001

Practical implication:
Select each component independently based on domain-specific needs

Metric Insights

- **LSE-C/D** primarily reflect audio, not visual sync
  - Consistent with Zhang et al. (ICIP 2024)
  - SyncNet bias documented by Yaman et al. (ECCV 2024)
- **AVSu** showed no sensitivity to any factor
- **SSIM** is the most discriminative visual metric (η^2 = 0.85)

Comparison with Literature

Finding	Prior work	Our result
LSE-C/D correlate with sync quality	Zhang (ICIP '24): **poor correlation**	Confirmed: LSE-C/D track VC choice, not lipsync
SyncNet biased toward neutral	Yaman (ECCV '24): yes	No emotion effect at all in our metrics
Emotional speech degrades VC	ClonEval: partial degradation	**No effect** in our 5s clips
Combined pipeline = interaction	AV-Deepfake1M++: implied	**Largely additive** at metric level

The factorial design reveals what component-level benchmarks cannot: **independence, not interaction, is the dominant pattern.**

Limitations & Future Work

Limitations

- **Small pool:** 5 actors, Spanish only
- **MuseTalk disabled** (Python 3.13 / mmpose)
- **8 SadTalker failures** (non-random)
- **High WER** across all systems (mean 1.12)
- **Proxy metrics**, not full SyncNet/AV-HuBERT
- **No human evaluation** yet

Next Steps

- PLACEHOLDER **Human evaluation** (N=30, 4 MOS dimensions)
  - Design ready, Flask app built
- Include MuseTalk when mmpose updated
- Test additional languages
- Longer clips (>5s)
- Full metric implementations
- VC × Lipsync interaction in human perception

Human Evaluation Design PLANNED

Protocol

- **N = 30** participants
- 2 identities per participant
- 32 conditions each (4 VC × 4 LS × 2 emo)
- **64 ratings** per participant

4 MOS Dimensions (1–5)

1. Overall quality
2. Lip synchronization
3. Voice naturalness
4. Visual naturalness

Statistical Analysis

LMM:

$$\text{MOS} \sim \text{VC} \times \text{LS} \times \text{Emo} + (1|\text{part.}) + (1|\text{id})$$

H6: Metric validity

Spearman ρ between computational metrics and human MOS

- Bootstrap 95% CI (10k iterations)
- Compare with THEval's $\rho = 0.870$

H1–H3: Human Ratings by System PLACEHOLDER

MOS by VC System (H1)

VC System	Overall	Lip sync	Voice nat.	Visual nat.
XTTS-v2	—	—	—	—
kNN-VC	—	—	—	—
OpenVoice V2	—	—	—	—
CosyVoice 2	—	—	—	—

MOS by Lipsync System (H2)

Lipsync	Overall	Lip sync	Voice nat.	Visual nat.
Wav2Lip	—	—	—	—
SadTalker	—	—	—	—
VideoReTalking	—	—	—	—

LMM Results (H1–H3)

Effect	MOS dim.	F	p	η_p^2
VC (H1)	Overall	—	—	—
VC (H1)	Voice nat.	—	—	—
LS (H2)	Overall	—	—	—
LS (H2)	Lip sync	—	—	—
VC×LS (H3)	Overall	—	—	—
VC×LS (H3)	Lip sync	—	—	—

> Fill with LMM output after running `05_run_analysis.py` on human data

H4–H5: Emotion in Human Perception PLACEHOLDER

MOS by Emotion (H4)

Condition	Overall	Lip sync	Voice nat.	Visual nat.
Neutral	—	—	—	—
Emotional	—	—	—	—
Difference	—	—	—	—

LMM: Emotion Effects

Effect	MOS dim.	F	p	η_p^2
Emotion (H4)	Overall	—	—	—
Emo×VC (H5)	Overall	—	—	—
Emo×LS (H5)	Lip sync	—	—	—

Key Question

Do human judges detect emotion effects that **computational metrics missed**?

- Computational: 0/10 metrics significant for emotion
- Human: PLACEHOLDER — / 4 MOS dimensions significant

> If humans detect differences that metrics don't → strong case for human evaluation in talking head benchmarks

H6: Metric Validity PLACEHOLDER

Spearman ρ : Computational Metrics vs. Human MOS

Metric	vs. Overall	vs. Lip sync	vs. Voice nat.	vs. Visual nat.
LSE-C	—	—	—	—
LSE-D	—	—	—	—
SSIM	—	—	—	—
LMD	—	—	—	—
CPBD	—	—	—	—
WavLM sim	—	—	—	—
Mel sim	—	—	—	—
WER	—	—	—	—

Bootstrap 95% CI (10k iterations). Compare with THEval's best: $\rho = 0.870$

[Insert scatter plot: best metric vs. human MOS — `figures/fig_metric_validity.png`]

Thank You

A Factorial Benchmark of Voice Cloning and Lip Synchronization Pipelines

4 VC × 3 Lipsync × 2 Emotions × 5 Actors = 232 stimulus videos

VC controls audio (η^2 = 0.73) | Lipsync controls visual (η^2 = 0.85) | Emotion = no effect

Code & data: PLACEHOLDER GitHub / OSF

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